

INDIVIDUAL PROJECT ASSIGNMENT - Java web programming

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Create an application that enables the online purchase of selected products. The application has the following characteristics/properties

- There are 2 roles in the application: user (customer) and administrator

USER

- An anonymous user can browse categories (**anonymous**)
- An anonymous user can view products (**anonymous**)
- User can add product to cart (**anonymous**)
- The user can define the quantity of a certain product that he puts in the basket (eg 20 protective foils for a mobile phone) (**anonymous**)
- The user can change the composition of the cart (e.g. completely remove a product or a certain quantity of a product from the basket or all products) (**anonymous**)
- The user can purchase products online; cash options are possible
 - cash on delivery and PayPal (all information about the implementation of PayPal, creating a sandbox account, etc., can be found at <https://developer.paypal.com/>) (**authenticated - customer**)
- The user must be able to see a historical overview of his purchases with an overview of the following data: (**authenticated - customer**)
 - What was purchased (contents of the basket, with quantities)
 - When it was purchased
 - How it was purchased (cash - cash on delivery or PayPal)

ADMINISTRATOR

- Administrator manages categories and products (CRUD operations on categories and products) (authenticated - admin)
- The administrator can see a historical overview of data about who logged in, when and from which IP address (authenticated - admin)
- The administrator can see a complete historical overview of purchases, for all customers, with the same information. Enable simple search filters by customer and time period of purchase (authenticated - admin)

GENERAL INSTRUCTIONS

- Visual design: Do your best (hint: Bootstrap)
- The application must be installed on some hosting before the presentation (mandatory for passing the exam). Running the application from the development environment is not enough to pass the course.
- Must use Spring MVC (LO1) + Thymeleaf (or equivalent template engine, e.g. JSP+JSTL, FreeMaker, Mustache..) (LO2 + LO3)
- Application design: application of good practices
- It is mandatory to find at least one suitable place where you will apply security filters (LO5) and listeners (LO6)
- Implement asynchronous mechanism for publishing data on the web page
- Implement authentication/authorization elements of the application according to the given specification using the Spring Security approach (LO4)
- Implement JWT tokens for REST API security
- Scan your application with SonarQube and resolve ALL reported issues (mandatory for passing the exam).